

Lec 10-2, Nov 12, Wed.

- Trigonometric integrals (8.3)

- $\int \sec^n x \, dx$

- $\int \tan^m x \sec^n x \, dx$

- Integrands with $\sqrt{1 \pm \cos 2\theta}$

- $\int \sin^m x \sin^n x \, dx$, $\int \sin^m x \cos^n x \, dx$, $\int \cos^m x \cos^n x \, dx$.

- Trigonometric substitutions (8.4)

Trigonometric Integrals.

$$\textcircled{1} \int \sin^m x \cos^n x \, dx$$

$$\textcircled{2} \int \tan^m x \, dx$$

$$\textcircled{3} \int \sec^n x \, dx \quad (n \in \mathbb{N}).$$

$$\textcircled{4} \int \tan^m x \sec^n x \, dx \left(= \int \frac{\sin^m x}{\cos^{m+n} x} \, dx \right), m, n \in \mathbb{Z}_+ \triangleq \{1, 2, 3, \dots\}.$$

1. If n is even, then take out a copy of $\sec^2 x$ and

express everything in terms of $\tan x$. This works even if $m=0$.

$$\left[\text{make use of } \begin{cases} \sec^2 x = \tan^2 x + 1 \\ d(\tan x) = \sec^2 x \, dx \end{cases} \right]$$

$$\int \tan^m x \sec^{\textcircled{2k}} x \, dx = \int \tan^m x \sec^{2k-2} x \sec^2 x \, dx$$

$$= \int \tan^m x \sec^{2k-2} x \, d \tan x$$

$$= \int \tan^m x (\tan^2 x + 1)^{k-1} \, d \tan x$$

$$\stackrel{y=\tan x}{=} \int y^m (y^2 + 1)^{k-1} \, dy \big|_{y=\tan x}.$$

[Numerical example : 8.4, example 7].

2. If m is odd, then take out a copy of $\tan x \sec x$,
and express the rest in terms of $\sec x$.

$$[\text{make use of } d(\sec x) = \frac{\sin x}{\cos^2 x} dx = \tan x \sec x dx]$$

$$\begin{aligned} \int \tan^{(2k+1)}(x) \sec^n(x) dx &= \int \tan^{2k}(x) \sec^{n-1}(x) \cdot (\tan x \cdot \sec x) dx \\ &= \int \tan^{2k}(x) \sec^{n-1}(x) d \sec x \\ &= \int (\sec^2 x - 1)^k \sec^{n-1}(x) d \sec x \\ &\stackrel{y=\sec x}{=} \int (y^2 - 1)^k y^{n-1} dy \Big|_{y=\sec x} \end{aligned}$$

3. If m is even and n is odd, then use $\tan^2 x = \sec^2 x - 1$

to reduce the problem to the case involving only $\int \sec^2(x) dx$.

$$\begin{aligned} \int \tan^{(2k)}(x) \sec^{(n)}(x) dx &= \int (\sec^2 x - 1)^k \sec^n(x) dx \\ &= \int [\sec^{2k} x - k \cdot (\sec^2 x)^{k-1} + \dots + (-1)^k] \sec^n(x) dx \\ &= \int \sec^{2k+n} x dx + \dots + (-1)^k \int \sec^n(x) dx. \end{aligned}$$

[Then use the reduction formula for $\int \sec^2(x) dx$].

• Integrands with $\sqrt{1 \pm \cos 2\theta}$.

Trigonometric identities involving squares, such as

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2} \quad \text{and} \quad \sin^2 \theta = \frac{1 - \cos 2\theta}{2},$$

may help evaluate integrals involving $\sqrt{1 \pm \cos 2\theta}$.

e.g. $\int_0^{\pi/4} \sqrt{1 + \cos 4x} \, dx = ?$

Sol: LHS = $\int_0^{\pi/4} \sqrt{1 + \cos(2 \cdot 2x)} \, dx$

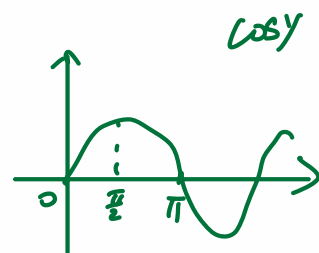
$$\begin{aligned} y &= 2x \\ \underline{\quad} \\ x &= \frac{y}{2} \end{aligned} \quad \int_0^{\pi/2} \sqrt{1 + \cos(2 \cdot y)} \, d\left(\frac{y}{2}\right)$$

$$= \frac{1}{2} \int_0^{\pi/2} \sqrt{1 + \cos 2y} \, dy$$

$$= \frac{1}{2} \int_0^{\pi/2} \sqrt{2 \cos^2 y} \, dy$$

$$= \frac{1}{2} \int_0^{\pi/2} \sqrt{2} \cos y \, dy \quad (\because \cos y \geq 0 \text{ on } [0, \frac{\pi}{2}])$$

$$= \frac{\sqrt{2}}{2} \sin y \Big|_0^{\pi/2} = \frac{\sqrt{2}}{2}.$$



□.

$$\cdot \int \sin(mx) \sin(nx) dx, \int \sin(mx) \cos(nx) dx, \int \cos(mx) \cos(nx) dx$$

These integrals can be evaluated by using integration by parts twice.

Alternatively, you may remember the following identities:

$$\sin mx \sin nx = \frac{1}{2} [\cos (m - n)x - \cos (m + n)x],$$

$$\sin mx \cos nx = \frac{1}{2} [\sin (m - n)x + \sin (m + n)x],$$

$$\cos mx \cos nx = \frac{1}{2} [\cos (m - n)x + \cos (m + n)x].$$

e.g. $\int \sin 4x \cos 8x dx = ?$

Sol : $\therefore \sin(4x+8x) = \sin(4x) \cos(8x) + \cos(4x) \sin(8x)$

$$\sin(4x-8x) = \sin(4x) \cos(8x) - \cos(4x) \sin(8x)$$

$$\therefore \sin 4x \cos 8x = \frac{1}{2} [\sin(4x+8x) + \sin(4x-8x)]$$

$$\therefore \text{LHS} = \int \frac{1}{2} [\sin(4x+8x) + \sin(4x-8x)] dx$$

$$= \frac{1}{2} \left[\int \sin(12x) dx + \int \sin(-4x) dx \right]$$

$$= \dots$$

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Trigonometric Substitution

- We start by considering $\int \frac{dx}{\sqrt{9+x^2}}$, with integrand defined on $\mathbb{R}$.

- If we consider $x = 3 \cdot \tan \theta$, then all x will be covered by considering all $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

- Hence, we may make the following substitution:

$$\begin{aligned} \int \frac{dx}{\sqrt{9+x^2}} & \stackrel{x=3\tan\theta}{=} \int \frac{1}{\sqrt{9+9\tan^2\theta}} d(3\tan\theta) \Big|_{x=3\tan\theta} \\ & = \int \frac{1}{3\sqrt{1+\tan^2\theta}} 3\sec^2\theta d\theta \Big|_{x=3\tan\theta} \\ & \stackrel{1+\tan^2\theta = \frac{\cos^2\theta+\sin^2\theta}{\cos^2\theta} = \sec^2\theta}{=} \int \frac{1}{\sqrt{1+\tan^2\theta}} \sec^2\theta d\theta \Big|_{x=3\tan\theta} \\ & \stackrel{\sec\theta > 0}{=} \int \frac{1}{\sqrt{\sec^2\theta}} \sec^2\theta d\theta \Big|_{x=3\tan\theta} \\ & = \int \sec\theta d\theta \Big|_{x=3\tan\theta} \\ & = \ln|\sec\theta + \tan\theta| + C \Big|_{x=3\tan\theta} \\ & = \ln\left|\sqrt{1+\frac{x^2}{9}} + \frac{x}{3}\right| + C \end{aligned}$$

Let $x = 3 \cdot \tan \theta$, $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$.
 $\therefore \cos \theta > 0 \Rightarrow \sec \theta > 0$
 $\therefore \begin{cases} \tan \theta = \frac{x}{3} \\ \cos^2 \theta + \sin^2 \theta = 1 \end{cases}$
 $\therefore \cos^2 \theta (1 + \tan^2 \theta) = 1$
 $= \cos^2 \theta (1 + \frac{x^2}{9}) = 1$
 $\therefore \cos \theta = \sqrt{1 + \frac{x^2}{9}} > 0$
 $\therefore \sec \theta = \sqrt{1 + \frac{x^2}{9}}$
 $\uparrow \uparrow$
Q. $\sec(\arctan(\frac{x}{3})) = ?$

[4]

The technique of trigonometric substitutions may be useful when integrating functions involving $\sqrt{a^2 - x^2}$, $\sqrt{x^2 - a^2}$ or $\sqrt{a^2 + x^2}$. The key idea is to make use the following trigonometric identities:

► $1 - \sin^2 \theta = \cos^2 \theta$.

► $\sec^2 \theta - 1 = \tan^2 \theta$.

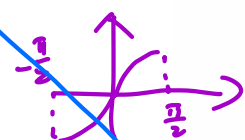
► $1 + \tan^2 \theta = \sec^2 \theta$.

$x = |a| \cdot \sin \theta$
 $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

$x = |a| \cdot \sec \theta$, $\theta \in [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$

$x = |a| \cdot \tan \theta$,
 $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

$-|a| \leq x \leq |a|$



$(\arctan \theta)' = \frac{1}{\theta^2 + 1}$

Depending on whether the integrand involves $\sqrt{a^2 - x^2}$, $\sqrt{x^2 - a^2}$ or $\sqrt{a^2 + x^2}$, we can make the substitution $x = a \sin \theta$, $x = a \sec \theta$ or $x = a \tan \theta$ accordingly.

Rmk: For the substitution $x = f(\theta)$, the domain of f is chosen so that f covers all the possible values of x .

e.g. Compute $\int_{-4}^{-3} \frac{1}{x^2 \sqrt{x^2 - 4}} dx$.

Sol: LHS $\stackrel{y=-x}{=} \int_3^4 \frac{1}{y^2 \sqrt{y^2 - 4}} dy$

$= \int_3^4 \frac{1}{y^2 \sqrt{\left(\frac{y}{2}\right)^2 - 1}} \cdot \frac{1}{2} dy$

$\stackrel{z=\frac{y}{2}}{=} \int_{\frac{3}{2}}^2 \frac{1}{z^2 \sqrt{z^2 - 1}} \cdot \frac{1}{4} dz$

$\therefore \sec : \theta \mapsto z$
 $(0, \frac{\pi}{2}) \rightarrow (1, \infty)$

Let $z = \sec \theta$, $\theta \in (0, \frac{\pi}{2})$.

$$\text{LHS } \frac{z = \sec \theta}{\theta = \operatorname{arcsec} z} \quad \frac{1}{4} \int_{\operatorname{arcsec} \frac{3}{2}}^{\operatorname{arcsec} 2} \frac{1}{\sec^2 \theta \sqrt{\sec^2 \theta - 1}} d(\sec \theta)$$

$$= \frac{1}{4} \int_{\operatorname{arcsec} \frac{3}{2}}^{\operatorname{arcsec} 2} \frac{1}{\sec^2 \theta \sqrt{\tan^2 \theta}} \tan \theta \sec \theta d\theta$$

$$\stackrel{\tan \theta > 0}{=} \frac{1}{4} \int_{\operatorname{arcsec} \frac{3}{2}}^{\operatorname{arcsec} 2} \frac{1}{\sec^2 \theta} \frac{\tan \theta \sec \theta}{\tan \theta} d\theta$$

$$= \frac{1}{4} \int_{\operatorname{arcsec} \frac{3}{2}}^{\operatorname{arcsec} 2} \cos \theta d\theta$$

$$= \frac{1}{4} \sin \theta \Big|_{\operatorname{arcsec} \frac{3}{2}}^{\operatorname{arcsec} 2}$$

$$= \frac{1}{4} \left[\sqrt{1 - \left(\frac{1}{2}\right)^2} - \sqrt{1 - \left(\frac{2}{3}\right)^2} \right]$$

$$= \frac{1}{4} \left[\frac{\sqrt{3}}{2} - \frac{\sqrt{5}}{3} \right]$$

$$\begin{aligned} \sec \theta_1 &= 2 \\ \Rightarrow \cos \theta_1 &= \frac{1}{2} \\ \Rightarrow \sin \theta_1 &= \sqrt{1 - \left(\frac{1}{2}\right)^2} > 0 \\ \text{---} \text{---} \text{---} \text{---} \text{---} \\ \sec \theta_2 &= \frac{3}{2} \\ \Rightarrow \cos \theta_2 &= \frac{2}{3} \\ \Rightarrow \sin \theta_2 &= \sqrt{1 - \left(\frac{2}{3}\right)^2} > 0 \end{aligned}$$

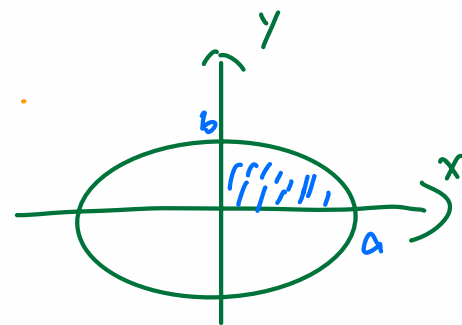
□

e.g. Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (where $a, b > 0$).

Sol: Area = $4A_1$, where A_1 is the shaded area.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Leftrightarrow y^2 = b^2 \left(1 - \frac{x^2}{a^2}\right)$$

$$\Rightarrow y = b \sqrt{1 - \frac{x^2}{a^2}}, \quad x \in [0, a] \text{ (on } A_1 \text{)}.$$



$$A_1 = \int_0^a y(x) dx = \int_0^a b \sqrt{1 - \frac{x^2}{a^2}} dx$$

$$\stackrel{\frac{x}{a} = z}{=} \int_0^1 b \sqrt{1 - z^2} d(az)$$

$$= ab \int_0^1 \sqrt{1-z^2} dz$$

$$\begin{array}{l} z = \sin \theta \\ \theta = \arcsin z \end{array} \quad ab \int_0^{\frac{\pi}{2}} \sqrt{1-\sin^2 \theta} d \sin \theta$$

$$= ab \int_0^{\frac{\pi}{2}} \cos \theta \cdot \cos \theta d\theta$$

$$= ab \int_0^{\frac{\pi}{2}} \frac{1+\cos 2\theta}{2} d\theta$$

$$= ab \left[\frac{1}{2} \cdot \theta + \frac{1}{4} \sin 2\theta \right] \Big|_0^{\frac{\pi}{2}}$$

$$= ab \cdot \frac{\pi}{4}.$$

$$\therefore \text{Area} = 4 A_1 = 4 \cdot \frac{1}{4} ab \pi = ab \pi.$$

□.

Particularly . if $a=b=r$, $\text{Area} = \pi r^2$.

• Integration by Partial Fractions

We now introduce a method that works well for integrating rational functions $P(x)/Q(x)$. Consider finding

i.e., P and Q are polynomials $\int \frac{5x-3}{x^2-2x-3} dx. \checkmark$

This becomes easy if we realize that

$$\frac{5x-3}{x^2-2x-3} = \frac{2}{x+1} + \frac{3}{x-3},$$

since the right-hand side is easy to integrate. How do we split $P(x)/Q(x)$ into rational functions that are easier to integrate?

$$\deg(3x^4 - 2x^2 + x - 100) = 4.$$

Def. If $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ with $a_n \neq 0$, then $\deg(P(x)) = n$. Here $\deg(P(x))$ is called the degree of $P(x)$.

[We do not consider the degree of a zero polynomial here.]

e.g. $\deg(7x^3 - 2x^2 + \pi x + e) = 3$;

$$\deg(a_0) = 0 \text{ (for } a_0 \neq 0 \text{)}.$$

Fact : If $P(x)$ and $Q(x)$ are polynomials with $Q(x)$ nonzero, then there exist polynomials $S(x)$ and $R(x)$ such that

(quotient) (remainder)

$$P(x) = Q(x) \cdot S(x) + R(x) ;$$

$$R(x) \equiv 0 \text{ or } \deg(R(x)) < \deg(Q(x)) .$$

This means that

$$\frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)} . \quad (*)$$